THE EULER SEQUENCE ON PROJECTIVE SPACE, WITH THE TWO MAPS WRITTEN OUT ON THE STANDARD OPENS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Projective space as Proj of a polynomial ring with its standard opens. Same chapter derham.tex, the discussion preceding the lemma (used at proof lines 1602-1617). It fixes $\mathbf{P}^n_A = \operatorname{Proj}(A[T_0, \ldots, T_n])$ and the standard opens $D_+(T_i) = \Spec(A[T_0/T_i, \ldots, T_n/T_i])$ on which the maps are written.
- The tautological quotient and the identifications of $\mathcal{O}(-1)$ and $\mathcal{O}$ on the standard opens. Same chapter derham.tex, the description of the relevant invertible sheaves used at proof lines 1606-1608 and 1624. They let $\mathcal{O}(-1)^{\oplus n+1}$ be written as $\bigoplus_j \mathcal{O}(-1)\mathrm{d}T_j$ and make sense of the coefficients $1/T_i$ and T_j appearing in the two maps.

Lemma 0.1. *There exists a short exact sequence*

$$0 \to \Omega \to \mathcal{O}(-1)^{\oplus n+1} \to \mathcal{O} \to 0$$

Proof. To explain this, we recall that $\mathbf{P}^n_A = \mathrm{Proj}(A[T_0, \ldots, T_n])$, and we write symbolically

$$\mathcal{O}(-1)^{\oplus n+1} = \bigoplus\nolimits_{j=0,\ldots,n} \mathcal{O}(-1)\mathrm{d}T_j$$

The first arrow

$$\Omega \to \bigoplus\nolimits_{j=0,\ldots,n} \mathcal{O}(-1)\mathrm{d}T_j$$

in the short exact sequence above is given on each of the standard opens $D_+(T_i) = \mathrm{Spec}(A[T_0/T_i, \ldots, T_n/T_i])$ mentioned above by the rule

$$\textstyle\sum_{j\neq i} g_j \mathrm{d}(T_j/T_i) \longmapsto \sum_{j\neq i} g_j/T_i \mathrm{d}T_j - (\sum_{j\neq i} g_j T_j/T_i^2)\mathrm{d}T_i$$

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This makes sense because $1/T_i$ is a section of $\mathcal{O}(-1)$ over $D_+(T_i)$. The map

$$\bigoplus\nolimits_{j=0,\ldots,n} \mathcal{O}(-1)\mathrm{d}T_j \to \mathcal{O}$$

is given by sending $\mathrm{d}T_j$ to T_j, more precisely, on $D_+(T_i)$ we send the section $\sum g_j \mathrm{d}T_j$ to $\sum T_j g_j$. We omit the verification that this produces a short exact sequence. $\square$